GenAI

NON-WSQ · 2 DAYS

COURSE SLIDES · NON-WSQ

Business Transformation with AI Agents

Course Code: C997

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

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About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE

Principal Trainer, Tertiary Infotech Academy Pte Ltd

CERTIFICATION

Generative AI, digital transformation and data analytics — 20+ years in technology and innovation.

DELIVERS

Courses on generative AI, business innovation, automation and data science.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

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ICE-BREAKER

Let's Know Each Other

- Your name, organisation and role.
- Which generative AI tools you already use, and what for.
- One business problem in your organisation you would like GenAI to solve.
- What you want to be able to do by the end of Day 2.

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HOUSEKEEPING

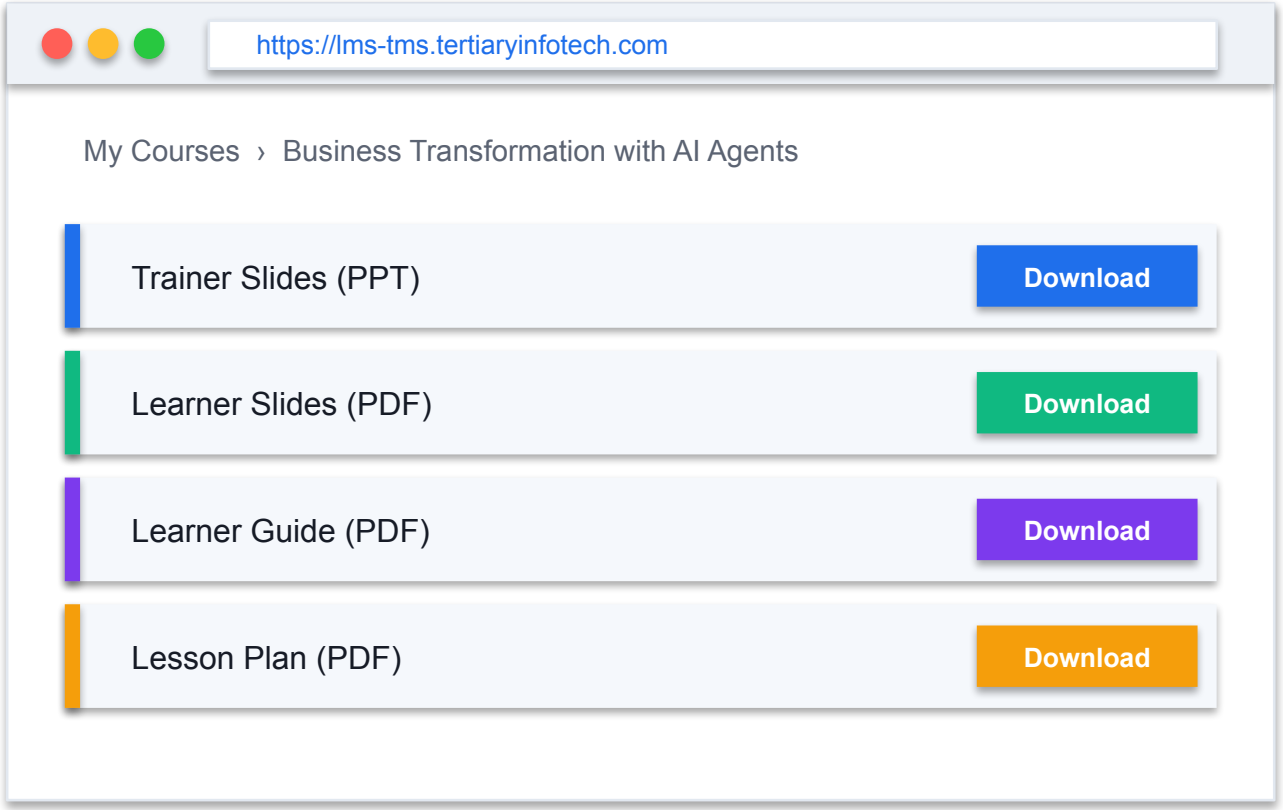
Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. Attend both days in full to get the most from the course.

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COURSE PORTAL · LMS / TMS

Download Your Course Material



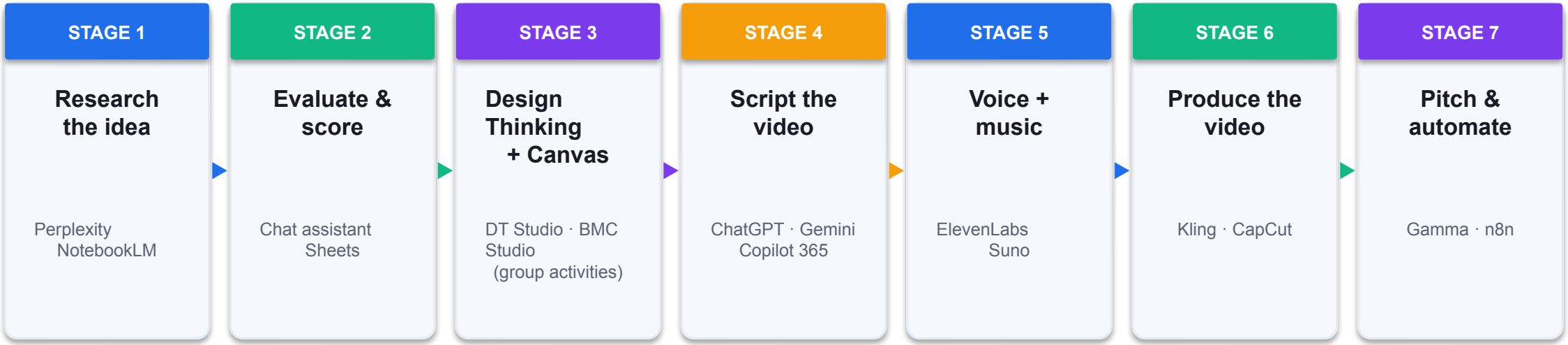
1. Sign in to lms-tms.tertiaryinfotech.com with the email you registered with.
2. Open My Courses and select this course.
3. Download the slides, Learner Guide and Lesson Plan.
4. Keep them open — you will refer to them in every lab.

Portal: https://lms-tms.tertiaryinfotech.com

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THE JOURNEY · STAGES 1–7

The Course Pipeline — One Idea, End to End



You choose ONE business idea in Lab 3 and carry it through every lab to the end of the course. Each lab consumes what the previous one produced — nothing is rebuilt from scratch. The course finishes by automating this entire pipeline as an agentic workflow with a human review gate.

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WHAT YOU'LL BUILD

Eighteen Connected Labs

1	**Predictive vs Generative AI** Same data, two framings	10	**AI Soundtrack** Suno, licence checked
2	**Map the GenAI Tool Landscape** Four modalities, hands-on	11	**Produce the Video** Kling + CapCut, captioned
3	**Research Your Idea** Perplexity + NotebookLM	12	**Business Presentation** Gamma, video embedded
4	**Evaluate & Prioritise** Weighted scoring rubric	13	**Risk-Assess the Pipeline** Your own assets
5	**Design Thinking Sprint** GROUP · DT Studio	14	**Guardrails & AI Ethics** 8 principles, 4 guardrail types
6	**Business Model Canvas** GROUP · BMC Studio	15	**Action Plan** 90 days, go/no-go
7	**Digital Architecture** Five layers + governance	16	**Measure Success** Baseline + counter-metric
8	**Script the Video** Three assistants compared	17	**Stakeholder Comms** Map, policy, idea intake
9	**AI Voiceover** ElevenLabs, 60 seconds	18	**CAPSTONE — Agentic Workflow** Automated, human-reviewed

Eighteen labs, each reusing the one before it. Nothing is rebuilt from scratch.

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ALL FREE TIERS · SIGN UP BEFORE LAB 1 (1 OF 2)

Your AI Toolkit — Research, Thinking & Text

Perplexity

www.perplexity.ai

Research with cited sources · Lab 3

NotebookLM

notebooklm.google.com

Reason over your documents · Lab 3

ChatGPT

chat.openai.com

Drafting and challenge · Labs 1, 4, 8

Google Gemini

gemini.google.com

Current web context · Lab 8

Copilot 365

copilot.microsoft.com

Business documents · Lab 8

DT Studio

alfredang.github.io/designthinking

Design Thinking · Lab 5

BMC Studio

alfredang.github.io/bcm

Business Model Canvas · Lab 6

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ALL FREE TIERS · SIGN UP BEFORE LAB 1 (2 OF 2)

Your AI Toolkit — Voice, Video & Automation

ElevenLabs

elevenlabs.io

Text-to-speech voiceover · Lab 9

Suno

suno.com

AI background music · Lab 10

Kling AI

klingai.com

Text-to-video scenes · Lab 11

CapCut

www.capcut.com

Edit, caption, export · Lab 11

Gamma

gamma.app

AI presentations · Lab 12

n8n

n8n.io

Agentic workflows · Lab 18

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THREE TOPICS · 18 CONNECTED LABS

Course Outline

Topic 1

- Predictive vs generative AI
- The GenAI tool landscape
- Business value of GenAI
- Investigating opportunities
- Evaluating and prioritising

Topic 2

- Business Model Canvas
- Design Thinking with GenAI
- Digital architecture design
- Integrating GenAI tools
- Building the action plan

Topic 3

- Implementation best practices
- Legal, ethical & security issues
- Measuring and reviewing success
- Stakeholder communication
- The AIM framework

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SCHEDULE

Lesson Plan — 2 Days, 7.5 hours/day

Day 1

- Day 1 — Understanding GenAI and evaluating business opportunities

– Introductions · Outcomes · Course outline

– Topic 1: Introduction to Business Innovation and GenAI

– Labs 1–4: compare AI types, tool map, research your idea, score it

– Topic 2 begins: Design Thinking and the Business Model Canvas

– Labs 5–7: DT Studio sprint, BMC Studio canvas, digital architecture

Day 2

- Day 2 — Designing, implementing and governing GenAI innovation

– Topic 2 continues: the content production pipeline

– Labs 8–12: script, voiceover, soundtrack, video, presentation

– Topic 3: Implementing, governing and automating

– Labs 13–18: risk, guardrails & ethics, plan, KPIs, comms, CAPSTONE

– Course summary · Q&A · Course feedback

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01

TOPIC 01

Introduction to Business Innovation and Generative AI

Predictive vs generative AI · GenAI tool landscape · evaluating business opportunities

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LO1 · WHAT THIS TOPIC COVERS

Key Concepts — Topic 01

1

Predictive vs generative AI

Predictive AI scores and forecasts from historical data; generative AI creates new text, image, audio, video and code from a prompt.

2

How GenAI works

A foundation model is pre-trained on a large corpus, then aligned; at inference a prompt is turned into tokens and completed one token at a time.

3

Market momentum

The generative AI market is growing at a >35% CAGR; adoption has moved from experiment to core business process in under three years.

4

Where value comes from

Four value levers: revenue growth, cost and productivity, risk mitigation and sustainability.

5

The GenAI tool landscape

Text, image, speech and voice, music, video, digital human, 3D, office productivity, data analysis and IP tooling.

6

Evaluating an opportunity

Screen each idea for business value, feasibility, data readiness, risk and fit with organisational objectives before committing.

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WHY NOW

Generative AI — The Market Opportunity

$1.3T

Projected generative AI market by 2032

>35%

Compound annual growth rate

70%

Of organisations piloting or deploying GenAI

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THE SHIFT

Predictive AI vs Generative AI

PREDICTIVE AI

- Scores, ranks and forecasts.
- Learns from labelled historical data.
- Output: a number, class or probability.
- Churn scoring, demand forecast, fraud detection.
- Mature — in production for a decade.

GENERATIVE AI

- Creates new content.
- Learns patterns from very large corpora.
- Output: text, image, audio, video, code.
- Drafting, design, summarising, conversation.
- Fast-moving — new capability every quarter.

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UNDER THE HOOD

How Generative AI Works

1. Pre-train a foundation model on a very large corpus
2. Align it with human feedback so answers are useful and safe
3. Learner writes a prompt describing the task and context
4. Model predicts the most likely next token, repeatedly
5. Output is generated, then reviewed by a human

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WHERE THE VALUE COMES FROM

The Four Business Value Levers

1

Revenue growth

Faster product development, new revenue channels, personalised offers at scale.

2

Cost & productivity

Worker augmentation in drafting, summarising, coding and support; long-term talent optimisation.

3

Risk mitigation

Broader, deeper visibility of data improves pattern recognition and early risk detection.

4

Sustainability

Supports regulatory compliance and embeds sustainability into design and decisions.

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TEN MODALITIES

The Generative AI Tool Landscape

1. **Text**
   ChatGPT · Gemini · Claude · Copilot · Perplexity
2. **Image**
   Firefly · Designer · Canva · Leonardo · Midjourney
3. **Speech & voice**
   ElevenLabs · Descript · TTS and voice cloning
4. **Music**
   Suno · Udio · AIVA · Soundraw
5. **Video**
   Sora · Runway · Kling · CapCut · Veed
6. **Digital human**
   HeyGen · D-ID · Synthesia · Descript
7. **3D models**
   Luma AI · Meshy · Spline · Tripo
8. **Office productivity**
   Microsoft 365 Copilot · Google Workspace Gemini
9. **Data & analysis**
   Code interpreter · Copilot in Excel · BI assistants
10. **IP & legal**
    Patent and trademark search · contract review assistants

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REAL-WORLD ADOPTION

Use Cases Across Industries

1

Food & beverage

McCormick used AI with IBM to develop new flavour combinations under its ONE platform.

2

Media & entertainment

Script development, storyboarding, dubbing, de-ageing and localised content at scale.

3

Public safety

Analysing historical incident data to allocate resources proactively — with strict oversight.

4

Financial services

Client reporting, KYC summarisation, adviser copilots and fraud narrative generation.

5

Healthcare

Clinical note-taking, patient communication drafting and research summarisation.

6

Retail & e-commerce

Product descriptions, personalised campaigns and conversational shopping assistants.

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THE SCORING RUBRIC

Evaluating a GenAI Opportunity

1

Business value

How much revenue, cost or risk does it actually move?

2

Feasibility

Can it be built and adopted with the capability we have?

3

Data readiness

Does the data exist, and are we permitted to use it?

4

Risk

What is the legal, ethical, security and reputational exposure?

5

Strategic fit

Does it advance a stated organisational objective?

6

Decision

Score, weight, rank — then defend the top two and the discards.

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WHAT YOU'LL DO

Hands-On Labs — Topic 01

Labs 1–2

- Compare Predictive and Generative AI on a Business Task
- Map the Generative AI Tool Landscape

Lab 3

- Research a Business Idea with Perplexity and NotebookLM

Lab 4

- Evaluate and Prioritise Your Business Idea

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TOPIC 01 · HANDS-ON

Compare Predictive and Generative AI on a Business Task

LAB 1

Run the same customer-feedback data through a predictive framing and a generative framing, then contrast what each produces and when a business should reach for which.

You'll build

A one-page comparison table showing predictive vs generative output on the same business problem, with a recommendation on when to use each.

Tools: ChatGPT, Google Gemini or Claude (free tier)

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LAB 1 · VERIFY

Compare Predictive and Generative AI on a Business Task

✅ Test it

Your table clearly shows predictive AI producing a label or forecast from historical data, and generative AI producing new content — and you can justify which one fits a named task in your organisation.

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TOPIC 01 · HANDS-ON

Map the Generative AI Tool Landscape

LAB 2

Build a personal tool map by testing one tool in each of four modalities, so you can pick the right tool for a business need instead of defaulting to a chat assistant. These are the tools you will use for the rest of the course.

You'll build

A tool map listing at least eight GenAI tools by modality, each with its business use case, pricing tier and one limitation you observed.

Tools: ChatGPT/Gemini/Claude, Perplexity, Adobe Firefly or Microsoft Designer, Suno, Canva

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LAB 2 · VERIFY

Map the Generative AI Tool Landscape

✅ Test it

Your tool map covers at least four modalities you tested first-hand plus three more from the slides, and for each tool you can name the business task it best supports and one limitation.

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TOPIC 01 · HANDS-ON

Research a Business Idea with Perplexity and NotebookLM

LAB 3

STAGE 1 OF THE PIPELINE. Choose the business idea you will carry through the whole course, then research it properly: Perplexity for the external market with cited sources, NotebookLM for synthesising the documents into a grounded briefing you can trust.

You'll build

A research briefing on ONE chosen business idea: market size and trend, three competitors, the customer problem, and three verified sources — plus a NotebookLM notebook containing the underlying documents.

Tools: Perplexity (Deep Research), Google NotebookLM, ChatGPT/Gemini/Claude

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LAB 3 · VERIFY

Research a Business Idea with Perplexity and NotebookLM

✅ Test it

Your briefing fits one page, every statistic traces to a source you personally opened, and your NotebookLM notebook answers the core-problem question with an inline citation.

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TOPIC 01 · HANDS-ON

Evaluate and Prioritise Your Business Idea

LAB 4

STAGE 2 OF THE PIPELINE. Stress-test the idea from Lab 3 against alternatives using an explicit scoring rubric, so the decision to proceed can be defended to management rather than being a matter of taste.

You'll build

A scored opportunity register comparing your chosen idea against five alternatives on business value, feasibility, data readiness, risk and strategic fit — with a defended recommendation.

Tools: ChatGPT/Gemini/Claude, Excel or Google Sheets

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LAB 4 · VERIFY

Evaluate and Prioritise Your Business Idea

✅ Test it

Your register scores all six ideas on the five weighted criteria, your chosen idea maps to a named organisational objective, and you can explain why the lowest-ranked idea was rejected.

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TOPIC RECAP

Recap — Topic 01

1. Distinguish current predictive ai from emerging generative ai and describe where each creates business value.
2. Survey generative ai tools across text, image, audio, video, digital human, productivity and data analysis.
3. Investigate business strategies, and the organisation's operating context, priorities, domain and environment, to identify business opportunities created by generative ai.
4. Evaluate opportunities for viability, applicability to the organisation and compatibility with business goals.

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02

TOPIC 02

Designing Digital Architectures and Processes for Business Innovation

Business Model Canvas · Design Thinking · digital architecture · action planning

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LO2 · WHAT THIS TOPIC COVERS

Key Concepts — Topic 02

1

Business planning methodologies

The Business Model Canvas, Value Proposition Canvas and Design Thinking give innovation a repeatable structure.

2

AI-powered Business Model Canvas

Use GenAI to draft and stress-test each of the nine blocks, then validate the assumptions with evidence.

3

Design Thinking with GenAI

Empathise, Define, Ideate, Prototype and Test — GenAI accelerates every stage, especially ideation and prototyping.

4

Digital architecture layers

Data, model, orchestration, application and governance layers describe where AI sits in the business.

5

Integration patterns

Chat assistant, RAG over company documents, workflow automation and autonomous agents are the four common patterns.

6

From design to action plan

An action plan names the owner, milestone, resource, risk and success measure for every initiative.

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STRUCTURE FOR INNOVATION

Business Planning Methodologies for Innovation

1

Business Model Canvas

Nine blocks describing how the organisation creates, delivers and captures value.

2

Value Proposition Canvas

Matches customer jobs, pains and gains to your products and services.

3

Design Thinking

Empathise, Define, Ideate, Prototype, Test — human-centred and iterative.

4

Lean experimentation

Identify the riskiest assumption and test it cheaply before committing.

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GROUP ACTIVITIES · LABS 5–6

BMC Studio — Your Canvas Ed-Tool

Design Thinking Studio

alfredang.github.io/designthinking

5 stages · empathy map · voting — Lab 5

BMC Studio

alfredang.github.io/bcm

Nine blocks · Strategyzer prompts — Lab 6

One owner creates

Share the Canvas ID

Team joins by link or QR code

Everyone edits live

Real-time sticky notes

See teammates' initials as they work

Nine blocks guided

Strategyzer prompts built in

Each block asks the right question

Tag and prioritise

Category · priority · status

Vote on the riskiest assumptions

Capture for Lab 12

Screenshot or share link

Goes straight into your pitch deck

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CANVAS GUIDELINES (1 OF 2)

Six Rules for a Good Business Model Canvas

1 Be specific

'Busy working parents in Singapore with children aged 4–8' beats 'families'. A vague segment produces a vague everything-else.

2 Make it testable

If no evidence could disprove a block, it is a slogan, not a business model. Write what would prove you wrong.

3 Keep it consistent

Revenue must come from the segment you named, through the channel you listed, using the resources you have.

4 3–5 items per block

A canvas crammed with everything says nothing. Force the prioritisation.

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CANVAS GUIDELINES (2 OF 2)

Six Rules for a Good Business Model Canvas

1 Customer language

Write what the customer would say, not internal jargon. If they would not recognise the pain, you have not found it.

2 Hypothesis, not plan

Mark what is ASSUMED versus EVIDENCED. Then test the three riskiest assumptions within two weeks.

THE MOST COMMON FAILURE

Accepting the AI's first draft. A chat assistant writes a confident, generic canvas in seconds — it reads well and says nothing specific to your market. Feed it your Lab 3 research, then REWRITE at least four blocks with what only you know. The canvas is where your research becomes a business model, not where the AI guesses one.

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NINE BLOCKS, AI-ASSISTED

The AI-Powered Business Model Canvas

1

Customer segments

Cluster and profile segments from your own data.

2

Value propositions

Draft and stress-test alternative propositions fast.

3

Channels

Generate channel-specific content at scale.

4

Customer relationships

Assistants and personalised follow-up at every touchpoint.

5

Revenue streams

Model pricing scenarios and bundle options.

6

Key resources

Identify the data, models and skills actually required.

7

Key activities

Automate drafting, research, analysis and reporting.

8

Key partners

Map vendors, model providers and integration partners.

9

Cost structure

Estimate licence, inference, integration and change costs.

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HUMAN-CENTRED, AI-ACCELERATED

Design Thinking with Generative AI

1. EMPATHISE — draft personas and interview guides, then talk to real users
2. DEFINE — converge on one 'How might we' problem statement
3. IDEATE — diverge to twenty ideas, then converge to three
4. PROTOTYPE — generate concepts, copy, visuals and mockups quickly
5. TEST — put it in front of users and measure what actually happens

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FIVE LAYERS

Digital Architecture for a GenAI Solution

APPLICATION	Where the user works — chat window, intranet, Teams, or an existing business system.
ORCHESTRATION	Prompts, retrieval, workflow automation and agent logic that connect data to the model.
MODEL	The foundation model or AI service — and whether your data leaves your tenancy when it is called.
DATA	Source systems, documents and records — with owner, sensitivity and permission recorded.
GOVERNANCE	Access control, human review points, logging, audit and prohibited-data rules across all layers.

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CHOOSING THE PATTERN

Four Integration Patterns

1

Chat assistant

A general assistant for staff productivity. Fastest to adopt, lowest integration cost.

2

RAG over documents

Grounds answers in your own content so responses cite the organisation's truth.

3

Workflow automation

GenAI embedded as a step inside an existing business process.

4

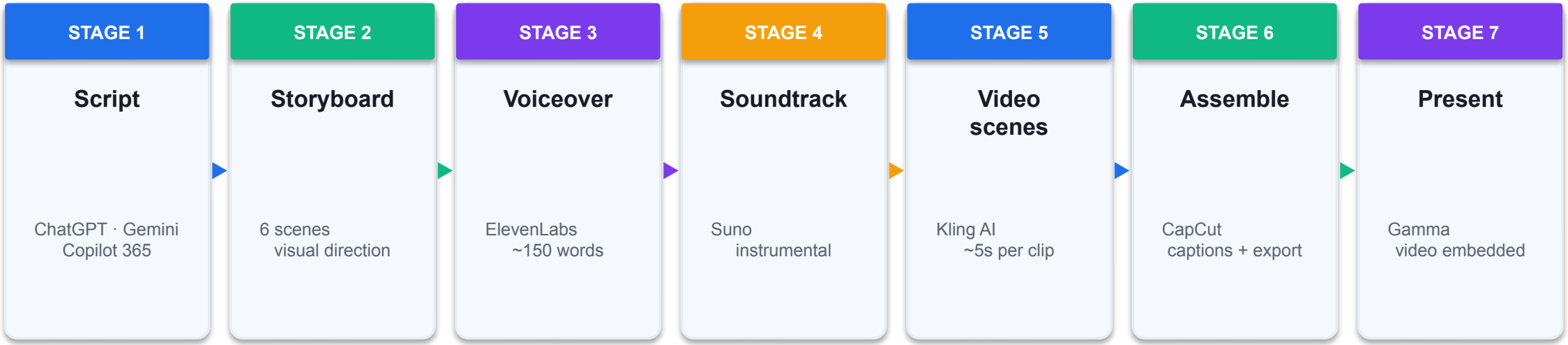
Autonomous agent

Plans and executes multi-step tasks with tools — highest value, highest oversight need.

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LABS 8–12

The Content Production Pipeline



Each stage consumes the previous one's output. The script sets the pace of the whole video — get it to 150 words and 60 seconds before you generate a single asset.

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TEXT TOOLS · LAB 8

Prompting the Three Text Assistants

WHAT EACH IS BEST AT

- ChatGPT — most creative; strongest hooks and taglines.
- Gemini — most current; can pull live web context.
- Copilot 365 — most business-appropriate; reads your own M365 documents.
- Run the SAME brief through all three, then combine the best parts.

WHAT MAKES A PROMPT WORK

- Give it a ROLE: 'You are a marketing copywriter.'
- State the AUDIENCE, the ONE message and the desired action.
- Specify FORMAT and LENGTH: '60 seconds, mark the timing.'
- Ask it to CHALLENGE you — assistants agree far too readily.

150 words ≈ 60 seconds of natural speech. Read the script aloud with a stopwatch before you go further.

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AUDIO & VIDEO TOOLS · LABS 9–11

Voice, Music and Video — The Craft Tips

1 Voiceover · ElevenLabs

Punctuation IS direction: a comma is a short pause, an ellipsis a beat. Write numbers and acronyms as they should be SAID ('twenty twenty-six', 'A-I'). Lower stability = more emotion, higher = steadier.

2 Music · Suno

Use Instrumental mode — lyrics fight the voiceover. Describe genre, mood, instruments and BPM. Never prompt 'in the style of <artist>'. Music sits 15–20 dB UNDER the voice.

3 Video · Kling

Name the subject, setting, camera move and lighting. Keep motion simple — hands, text and fast movement are where clips break. Reuse the same style words across scenes for consistency.

4 Assembly · CapCut

Lay the voiceover down FIRST, then cut visuals to it. Auto-caption, then PROOFREAD — 80% of social video is watched muted.

BEFORE YOU PUBLISH ANYTHING

Check the free-tier licence of every generated asset. Free tiers are frequently non-commercial or watermarked, and voice cloning requires written consent from the person whose voice it is.

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WHAT A VIABLE PLAN CONTAINS

From Design to Action Plan

1

Milestone

What is delivered, and by when.

2

Owner

A named role accountable for the outcome.

3

Resource

People, budget, licences and data access required.

4

Dependency

What must happen first, and who is blocking.

5

Risk & mitigation

What could derail it, and the control that prevents it.

6

Go / no-go

The numbers at week 12 that justify scaling, pausing or stopping.

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WHAT YOU'LL DO

Hands-On Labs — Topic 02

Labs 5–7

- GROUP ACTIVITY — Run a Design Thinking Sprint in Design Thinking Studio
- GROUP ACTIVITY — Build a Business Model Canvas in BMC Studio
- Design the Digital Architecture for Your Solution

Labs 8–10

- Script a Marketing Video for Your Business Idea
- Create the AI Voiceover
- Generate the Background Soundtrack with Suno

Labs 11–12

- Produce the Video with Kling and an AI Video Editor
- Build the Business Presentation with Your Video

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TOPIC 02 · HANDS-ON

GROUP ACTIVITY — Run a Design Thinking Sprint in Design Thinking Studio

LAB 5

Before modelling the business, understand the USER. In teams of 3-5, run all five Design Thinking stages on the researched idea using the course's Design Thinking Studio ed-tool, with AI accelerating each stage and real people correcting it.

You'll build

A completed five-stage Design Thinking workspace: an empathy map, one 'How might we' problem statement, twenty ideas narrowed to three, a prototype concept and a test plan — all captured in the shared studio.

Tools: Design Thinking Studio (https://alfredang.github.io/designthinking/), ChatGPT/Gemini/Claude, your Lab 3 NotebookLM notebook

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LAB 5 · VERIFY

GROUP ACTIVITY — Run a Design Thinking Sprint in Design Thinking Studio

✅ Test it

Your team's studio workspace has an artifact in all five stages, the empathy map was corrected by a real person's input rather than left as the AI's draft, your problem statement follows the 'How might we' form, and your test plan names a specific measurable signal that would disprove the concept.

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TOPIC 02 · HANDS-ON

GROUP ACTIVITY — Build a Business Model Canvas in BMC Studio

LAB 6

STAGE 3 OF THE PIPELINE, worked as a GROUP. In teams of 3-5, build a complete nine-block Business Model Canvas together in real time using the course's BMC Studio ed-tool, with AI drafting and challenging the content and the six canvas rules keeping it sharp.

You'll build

A collaborative nine-block Business Model Canvas in BMC Studio (shareable Canvas ID), annotated with where GenAI creates value in each block, plus the three riskiest assumptions and how the team would test them.

Tools: BMC Studio (https://alfredang.github.io/bcm/), ChatGPT/Gemini/Claude, your Lab 3 NotebookLM notebook

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LAB 6 · VERIFY

GROUP ACTIVITY — Build a Business Model Canvas in BMC Studio

✅ Test it

Your team's BMC Studio canvas has all nine blocks filled and mutually consistent, at least four blocks carry research-specific detail rather than the AI's generic draft, every block names a GenAI value-add, three high-priority assumptions are marked with a two-week test each, and every member contributed notes visible on the shared canvas.

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TOPIC 02 · HANDS-ON

Design the Digital Architecture for Your Solution

LAB 7

Draw the end-to-end architecture for the solution across five layers, so the design shows data flow, integration points and governance controls rather than just a list of tool names.

You'll build

A five-layer architecture diagram (data, model, orchestration, application, governance) with the integration pattern chosen and justified, and an explicit statement on whether confidential data leaves the organisation.

Tools: Draw.io, Canva or a chat assistant with diagram output

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LAB 7 · VERIFY

Design the Digital Architecture for Your Solution

✅ Test it

Your diagram shows all five layers with data-flow arrows, names at least one specific control in the governance layer, and states explicitly whether confidential data leaves the organisation's boundary.

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TOPIC 02 · HANDS-ON

Script a Marketing Video for Your Business Idea

LAB 8

STAGE 4 OF THE PIPELINE. Write and storyboard a 60-second video script for the researched idea, comparing how three different assistants handle the same creative brief.

You'll build

A finished 60-second video script with a hook, three benefits and a call to action, broken into a 6-scene storyboard with visual direction and the exact voiceover line for each scene.

Tools: ChatGPT, Google Gemini, Microsoft Copilot 365

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LAB 8 · VERIFY

Script a Marketing Video for Your Business Idea

✅ Test it

Your script reads aloud in 60 seconds or less, has a hook in the first five seconds, three clear benefits and one call to action, and your 6-scene storyboard gives each scene a voiceover line, a camera-ready visual description and on-screen text.

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TOPIC 02 · HANDS-ON

Create the AI Voiceover

LAB 9

STAGE 5 OF THE PIPELINE. Turn the Lab 8 script into a natural-sounding voiceover with a text-to-speech tool, and learn the controls that separate a usable take from an obviously robotic one.

You'll build

A clean 60-second voiceover audio file (MP3/WAV) of your script, with the voice, pace and emphasis deliberately chosen for your audience.

Tools: ElevenLabs (recommended free tier), Microsoft Edge Read Aloud, Descript Overdub, or Google/Azure TTS

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LAB 9 · VERIFY

Create the AI Voiceover

✅ Test it

Your voiceover plays cleanly, runs to 60 seconds or under, pronounces every product name and number correctly, and sounds natural when you play it to another learner.

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TOPIC 02 · HANDS-ON

Generate the Background Soundtrack with Suno

LAB 10

STAGE 6 OF THE PIPELINE. Create an original background track that supports the voiceover instead of competing with it, and check what the licence actually permits.

You'll build

A 60-second background music track matched to the tone of your video, with its licence terms recorded.

Tools: Suno (recommended), Udio, or an AI music generator of your choice

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LAB 10 · VERIFY

Generate the Background Soundtrack with Suno

✅ Test it

You have a 60-second instrumental track that fits your video's tone, sits comfortably under the voiceover when played together, and you can state exactly what its licence permits.

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TOPIC 02 · HANDS-ON

Produce the Video with Kling and an AI Video Editor

LAB 11

STAGE 7 OF THE PIPELINE. Generate the visual scenes from the Lab 8 storyboard, then assemble them with the Lab 9 voiceover and the Lab 10 soundtrack into a finished, captioned marketing video.

You'll build

A finished 60-second marketing video combining AI-generated scenes, your AI voiceover, your AI soundtrack and captions — exported at 1080p.

Tools: Kling AI (or Runway, Pika, Sora), CapCut or Descript, plus your Lab 9 and Lab 10 audio

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LAB 11 · VERIFY

Produce the Video with Kling and an AI Video Editor

✅ Test it

Your exported video plays end to end at 1080p with synced visuals, an audible voiceover over quieter music, accurate captions and a clear call to action — and you can state which AI tool produced each element.

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TOPIC 02 · HANDS-ON

Build the Business Presentation with Your Video

LAB 12

Assemble everything from the pipeline — research, canvas, architecture and video — into a board-ready presentation that argues for the idea rather than merely describing it.

You'll build

A published 10-slide business presentation with a share link, embedding your Lab 11 video and your Lab 6 canvas, ending with a clear ask.

Tools: Gamma (recommended), Canva Presentations, or Microsoft Copilot in PowerPoint

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LAB 12 · VERIFY

Build the Business Presentation with Your Video

✅ Test it

Your deck has a working share link, plays the embedded video, shows the canvas, carries at least one verified source from Lab 3, and ends with a specific, decidable ask.

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TOPIC RECAP

Recap — Topic 02

1	Apply a human-centred business planning methodology, accelerated by generative ai, to move from user problem to tested concept.
2	Apply a business planning methodology and design how genai supports each part of the business model.
3	Design a digital architecture showing how ai technologies apply across the business.
4	Chain generative ai tools into a repeatable content production process for the business.
5	Apply generative ai speech tools to produce a business-quality audio asset.
6	Apply generative ai music tools and handle the licensing of generated assets.
7	Combine multiple generative ai outputs into one finished business asset.
8	Turn the work into a stakeholder-ready business case and communicate it.

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03

TOPIC 03

Implementing and Managing Business Innovation with Generative AI Tools

Best practices · legal, ethical and security issues · measuring success · stakeholder communication

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LO3 · WHAT THIS TOPIC COVERS

Key Concepts — Topic 03

1

Implementation best practices

Start with a narrow high-value use case, keep a human in the loop, measure a baseline and scale only what works.

2

Legal considerations

Copyright and training data, model output ownership, the EU AI Act and Singapore's PDPA obligations.

3

AI ethics

Transparency, fairness, safety, explainability, privacy, security, responsibility, liability and auditability.

4

Singapore governance

IMDA's Model AI Governance Framework and AI Verify give organisations a practical assurance toolkit.

5

Security and data protection

Prompt injection, data leakage into public models, shadow AI and access control over AI systems.

6

Measuring and reviewing success

Track adoption, cycle time, quality, cost per task and business outcome against the pre-AI baseline.

7

Stakeholder communication

A communication plan, AI use policy and community of practice keep ideas flowing across the organisation.

8

The AIM framework

Assess, Implement, Manage — a repeatable loop for running generative AI innovation.

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WHAT ACTUALLY WORKS

Implementation Best Practices

1

Start narrow

One high-value use case with a measurable baseline beats ten pilots that prove nothing.

2

Human in the loop

Keep a review point wherever the output carries legal, financial or reputational weight.

3

Measure the baseline first

You cannot claim an improvement you never measured before the change.

4

Design for adoption

Train, communicate honestly about job impact, and make the tool easier than the old way.

5

Scale what works

Promote only the pilots that hit their go criteria; retire the rest without blame.

6

Iterate the loop

Assess, implement, manage — then reassess as models and regulation change.

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LEGAL

Legal & Copyright Considerations

THE CONTESTED GROUND

- Training data — Getty Images v Stability AI and related suits.
- Authorship — the US Copyright Office holds purely AI-generated work unprotectable.
- Likeness and voice cloning consent.
- AI-generated music and derivative-work claims.

WHAT TO DO

- Read the vendor terms on commercial use and ownership.
- Keep a human contribution record for creative work.
- Obtain written consent for any real person's likeness or voice.
- Disclose AI-generated content where regulation or policy requires.

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ETHICS

AI Ethics — The Principles

1. **Transparency**
   Users know when AI is used and how.
2. **Fairness**
   No unjustified bias against any group.
3. **Safety**
   The system behaves within intended bounds.
4. **Explainability**
   Decisions can be understood and justified.
5. **Privacy**
   Personal data is protected and minimised.
6. **Security**
   The system resists misuse and attack.
7. **Responsibility**
   A named human owns each AI outcome.
8. **Auditability**
   Decisions are logged and reviewable.

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GOVERNANCE IN SINGAPORE

Singapore's AI Governance Landscape

1

Model AI Governance Framework

IMDA's practical guidance on deploying AI responsibly, including for generative AI.

2

AI Verify

A testing framework and toolkit for validating AI system behaviour against governance principles.

3

PDPA

Consent, purpose limitation, retention and protection obligations for any personal data used.

4

Sector regulation

MAS, MOH and other regulators add sector-specific expectations on top of the baseline.

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SECURITY

Security Risks in GenAI Adoption

1

Prompt injection

Hostile instructions hidden in content the model reads, hijacking its behaviour.

2

Data leakage

Confidential material pasted into a public model and retained outside your control.

3

Shadow AI

Staff using unapproved tools with company data, invisible to IT and compliance.

4

Access control

Assistants inheriting broad permissions and exposing data users should not see.

5

Hallucination

Confident, fluent output that is simply wrong — dangerous without a review step.

6

Vendor & supply chain

Model, plugin and connector dependencies you do not control.

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THE KPI SET

Measuring and Reviewing Success

1

Adoption

Active users, tasks completed, share of the team using it weekly.

2

Efficiency

Cycle time and effort per task, against the pre-AI baseline.

3

Quality

Accuracy, rework rate and human-review pass rate.

4

Cost

Licence and inference cost per task versus the cost it replaced.

5

Business outcome

Revenue, retention, satisfaction or risk reduction actually moved.

6

Counter-metric

The measure that would reveal harm — errors, complaints, rework.

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Four Types of Guardrail

1 INPUT

What may never enter a prompt: personal data, customer records, unreleased financials, credentials, client-identifying material. Define which tools are approved for which data classification.

2 PROCESSING

Prompt-injection defence: treat all retrieved or uploaded content as UNTRUSTED, never let retrieved text change the system instruction, and restrict which tools an agent may call.

3 OUTPUT

Before anything is published: facts verified against a source, licence terms confirmed, brand and tone reviewed, AI disclosure applied where required.

4 HUMAN-IN-THE-LOOP

A named person approves wherever output is customer-facing, carries legal or financial weight, or cannot be reversed.

THE RULE THAT MATTERS MOST

Put the human gate BEFORE the expensive, irreversible steps. Approving the script costs nothing; discovering the problem after the video is generated and published costs credits, money and reputation.

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ETHICS PRINCIPLES · LAB 14

AI Ethics — From Principle to Practice

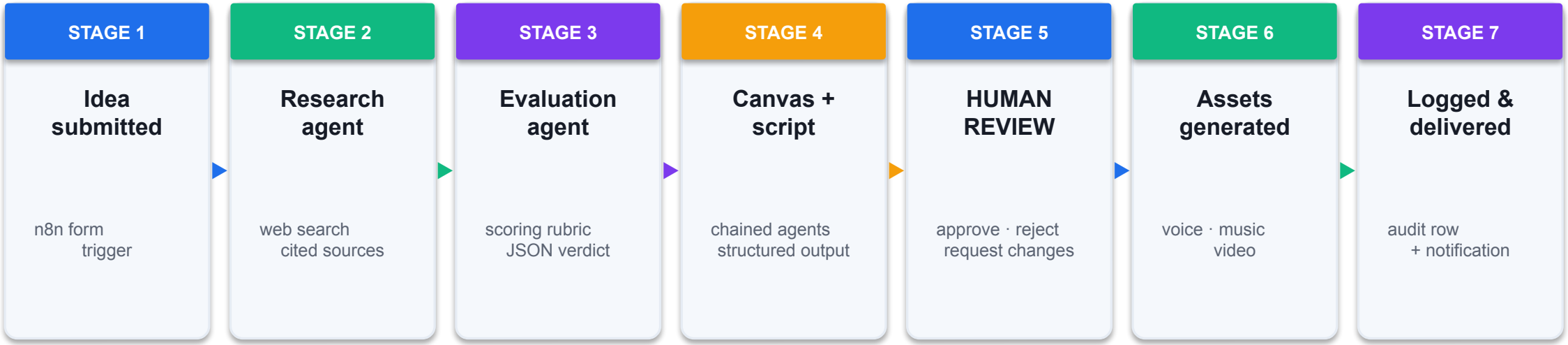
THE PRINCIPLE	WHAT IT LOOKS LIKE IN YOUR PIPELINE
Transparency — users know when AI is used.	Every AI-generated video carries an on-screen AI label.
Fairness — no unjustified bias against any group.	Run the same prompt with two demographic framings and compare.
Privacy — personal data protected and minimised.	No customer data ever enters a public tool. Ever.
Responsibility — a named human owns each outcome.	The approver's name is recorded on every published asset.
Auditability — decisions are logged and reviewable.	Every workflow run is logged: input, output, approver, time.

A principle without an observable behaviour is decoration. Write each one as something you could audit.

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LAB 18 · AUTOMATING THE WHOLE PIPELINE

The Capstone — An Agentic Workflow with Human Review



The workflow PAUSES at the human gate and only continues on approval — a rejection routes back to the script agent without spending a single video credit. A gate that cannot stop the process is not a gate.

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THE OPERATING LOOP

The AIM Framework for Implementing GenAI

1

ASSESS — investigate opportunities, evaluate viability and fit, pick the top use case

2

IMPLEMENT — design the architecture, plan the pilot, build with governance from day one

3

MANAGE — measure against baseline, review, communicate, and feed new ideas back in

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KEEPING IDEAS FLOWING

Facilitating Stakeholder Information Flow

1

Map the stakeholders

Plot influence against interest — sponsors, staff, IT, legal, customers.

2

Tailor the message

Executives need outcomes; affected staff need honesty about job impact.

3

Publish an AI use policy

Approved tools, prohibited data, disclosure rules and where to get help.

4

Build a community of practice

A channel and a monthly session where people share what worked.

5

Create an idea intake

A simple form so anyone can propose a use case.

6

Triage consistently

Score every submitted idea with the same rubric so the pipeline stays fair.

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WHAT YOU'LL DO

Hands-On Labs — Topic 03

Labs 13–14

- Risk-Assess Your Generative AI Pipeline
- Define Security Guardrails and AI Ethics Principles

Labs 15–16

- Develop the Implementation Action Plan
- Measure and Review Innovation Success

Labs 17–18

- Build the Stakeholder Communication and Idea-Sharing Plan
- CAPSTONE — Automate the Pipeline with an Agentic AI Workflow and Human Review

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TOPIC 03 · HANDS-ON

Risk-Assess Your Generative AI Pipeline

LAB 13

Run a structured risk assessment over the ACTUAL pipeline you built in Labs 3-12 — the research, canvas, script, voice, music, video and deck — and turn every finding into a named, owned control.

You'll build

A risk register covering your own pipeline stage by stage: copyright, data protection, accuracy, disclosure and security, each with a specific control and an owner.

Tools: Your Lab 3-11 artifacts, chat assistant, IMDA Model AI Governance Framework, PDPC guidelines

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LAB 13 · VERIFY

Risk-Assess Your Generative AI Pipeline

✅ Test it

Every pipeline stage has at least one identified risk with a specific owned control, your bias test is recorded with the actual prompts used, and you can state which of your generated assets are and are not safe to publish commercially.

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TOPIC 03 · HANDS-ON

Define Security Guardrails and AI Ethics Principles

LAB 14

Convert the risks found in Lab 13 into a working governance layer: the ethics principles the organisation commits to, and the concrete guardrails that enforce them in practice.

You'll build

A one-page AI governance charter: eight ethics principles with what each means in practice, plus a guardrail table mapping each risk to an input, processing, output or human control.

Tools: Chat assistant, IMDA Model AI Governance Framework, AI Verify, your Lab 13 risk register

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LAB 14 · VERIFY

Define Security Guardrails and AI Ethics Principles

✅ Test it

Every principle is written as an observable behaviour, every Lab 13 red or amber risk maps to a named guardrail with an owner, and your human-approval gates are specific about who approves what before publication.

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TOPIC 03 · HANDS-ON

Develop the Implementation Action Plan

LAB 15

Convert the design and its governance into a costed, owned and sequenced 90-day plan a management team could approve and track.

You'll build

A 90-day action plan with phased milestones, owners, resources, dependencies, a risk register and numeric go/no-go criteria.

Tools: Excel or Google Sheets, chat assistant

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LAB 15 · VERIFY

Develop the Implementation Action Plan

✅ Test it

Every milestone has a named owner role and date, the governance controls from Labs 13-14 appear as scheduled work, the risk register has five risks with mitigations, and the go/no-go criteria are numeric rather than subjective.

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TOPIC 03 · HANDS-ON

Measure and Review Innovation Success

LAB 16

Build the measurement system that shows whether the innovation actually worked, including the pre-AI baseline without which no improvement can be claimed.

You'll build

A KPI dashboard defining baseline, target and review cadence across adoption, efficiency, quality, cost and business outcome, plus a counter-metric.

Tools: Excel or Google Sheets, chat assistant

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LAB 16 · VERIFY

Measure and Review Innovation Success

✅ Test it

Every KPI has a real baseline number and a data source, at least one counter-metric is present, and your review cadence names who meets, how often and what decision they can make.

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TOPIC 03 · HANDS-ON

Build the Stakeholder Communication and Idea-Sharing Plan

LAB 17

Design how the innovation is communicated and how new ideas keep flowing after the pilot, so the initiative outlives its launch.

You'll build

A stakeholder map, a tailored communication plan, the published one-page AI use policy, and a mechanism for capturing and triaging new ideas.

Tools: Chat assistant, Gamma or Canva, Microsoft Teams or Slack

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LAB 17 · VERIFY

Build the Stakeholder Communication and Idea-Sharing Plan

✅ Test it

Every stakeholder group has a tailored message, channel, frequency and owner; the AI use policy fits one page and names prohibited data; and a submitted idea has a defined path to evaluation.

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TOPIC 03 · HANDS-ON

CAPSTONE — Automate the Pipeline with an Agentic AI Workflow and Human Review

LAB 18

The capstone. Everything you did by hand in Labs 3-12 becomes an automated agentic workflow — research, canvas, script, assets and deck — with a HUMAN REVIEW GATE before anything is published, and the Lab 14 guardrails enforced in the flow.

You'll build

A working agentic workflow that takes a business idea as input, runs the research-to-content pipeline through AI agents, pauses for human approval, and delivers the finished asset set — with the run logged for audit.

Tools: n8n (recommended), Make.com, Copilot Studio or Zapier; plus your Lab 3-11 tool accounts

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LAB 18 · VERIFY

CAPSTONE — Automate the Pipeline with an Agentic AI Workflow and Human Review

✅ Test it

Your workflow runs from a submitted idea through research, evaluation, canvas and script; it PAUSES at the human review gate and only continues on approval; a rejection routes back without generating assets; every run is logged with the approver and timestamp; and you can point to where each Lab 14 guardrail is enforced.

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TOPIC RECAP

Recap — Topic 03

1. Identify and address legal, ethical and security issues in implementing business innovation with genai.
2. Establish the ethical principles and technical guardrails that govern generative ai use across the organisation.
3. Apply best practices in the implementation process of business innovation, and develop a viable action plan in line with business strategy.
4. Manage business innovation and review the success of its integration with business strategy.
5. Facilitate information flow among key stakeholders to empower sharing and development of innovative ideas.
6. Design and build an agentic workflow that automates the business innovation process end to end, with human review and governance built in.

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WRAP-UP

Course Summary & Next Steps

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LEARNING OUTCOMES

What You Achieved

1

LO1 — Evaluated opportunities

Compared predictive and generative AI, mapped the tool landscape, researched real case studies and scored a prioritised opportunity register.

2

LO2 — Designed and planned

Built an AI-powered Business Model Canvas, ran a Design Thinking sprint, designed a five-layer digital architecture, shipped an AI assistant and a GenAI video.

3

LO3 — Implemented and managed

Produced a 90-day action plan, assessed legal/ethical/security risk, debated a real dilemma, built a KPI dashboard and a stakeholder communication plan.

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THE TAKEAWAY

Start narrow. Measure the baseline. Govern from day one.

The organisations that win with generative AI are not the ones with the most pilots — they are the ones that measured, governed and scaled the few that worked.

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NEXT STEPS

Your Next 30 Days

- Take your top-ranked opportunity from Lab 4 and socialise it with your sponsor.
- Measure the pre-AI baseline for that task before you change anything.
- Run the 90-day pilot plan from Lab 15 with the go/no-go criteria you defined.
- Publish the one-page AI use policy from Lab 17 so staff know what is permitted.
- Set up the idea intake and triage new use cases with the Lab 4 rubric.

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CONTINUE YOUR LEARNING

Recommended Follow-On Courses

1. Build Agentic AI and NLP Applications with Langflow
2. Build AI Agents and RAG Pipelines Without Code
3. Fundamentals of AI Ethics and Responsible AI
4. Mastering Prompt Engineering for Generative AI Content Creation
5. Build LLM Applications Using Langchain and Flowise

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IMPROVE Tell us what we could do better.

APPLY Share how you will use AI agents at work.

2 MIN Your feedback shapes the next run of this course.

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SEE YOU AT THE NEXT COURSE

Thank You!

You can now evaluate, design, implement and govern business innovation with generative AI in your own organisation.

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